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Phase-Field Modeling of Transport-Limited Electrolysis in Solid and Liquid States

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A phase-field model of electrochemical interface dynamics is developed to study cathode shape and topology change in transport-limited electrolysis in two and three dimensions under conditions of rapid charge redistribution. A case study for the binary model is carried out for an Fe–FeO system. Stability behavior of the model is in good agreement with linear stability theory for small amplitude sinusoidal perturbation in electrodeposition. When there is no convection, a high electric field and low surface tension cause the cathode interface to be unstable, leading to growth of dendrites which break into powders. When the electrodes and electrolyte are low-viscosity fluids, flow provides an additional mechanism for stabilizing the interface. A new stability criterion for this liquid situation based on the Schmidt number is derived from dimensional analysis and model results. For an unstable cathode interface, a streamer morphology (liquid dendrites) is observed in two and three dimensions. This binary model is extended to a ternary system and a representative case is carried out for the Ti–Mg–Cl system. One- and two-dimensional ternary simulations show qualitatively correct interface motion and electrical potential behavior.
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Electrochemical interactions are ubiquitous in the field of materials, from fuel cells and batteries, to smelting or refining of metals, to corrosion, to electromigration in electronic interconnects. In order to maximize the performance of these materials and systems, one must understand the role of electrochemistry in the migration of phase boundaries. The complexity of interface shape and topology changes and coupled chemical and electrical driving forces makes these systems very difficult to comprehend, let alone engineer. For example, the cathodic reduction of ions to metal exhibits a Mullins–Sekerka instability due to an ion-depleted layer at the cathode/electrolyte interface. Any perturbation on the cathode surface leads to the formation of solid dendrites or liquid streamers, affecting the process efficiency as well as resulting in a rough surface or porous deposit.

The governing equations which describe this process are well understood, such as the Poisson equation for electrical potential, charge conservation, and ion diffusion driven by electric and chemical potential gradients. Many investigators have used these equations in the limit of small perturbations in order to study the stability of cathodic interfaces.[1-9] Haataja et al. classify previous works on linear stability analysis into unsupported[1-8] and fully supported electrolytes,[3-5] then extend this analysis to arbitrary supporting electrolytes.[9] These analyses demonstrate that the interface perturbation growth is faster with increasing deposition rates, increasing spectator ion concentrations, increasing perturbation wavelengths and amplitudes, and decreasing ohmic resistances due to charge transfer. Qualitatively, these predictions correlate well with experimental observation.

The shape and topology of the interface, however, cannot be captured by linear stability analysis. In practice, an electrochemical process involves complex geometries and interacts with other physical phenomena such as fluid flow, solidification, stress state, etc. For these reasons, more general numerical models are required to investigate interface growth mechanisms. Methods that explicitly track the interface work sufficiently well under some circumstances[10] but break down when the interface topology changes. One particle-based numerical method that can accommodate interface topology changes is the kinetic Monte Carlo method. Martin has investigated the morphological stability of ion-conducting oxides and halides by kinetic Monte Carlo simulation.[11] Pricer et al. also use this method to model a copper electrodeposition process.[12] The limitation of the kinetic Monte Carlo model is that it is mainly based on phenomenological parameters in the probability density function, which must be carefully fitted to experimental data.

On the continuum side, one mean-field approach used to study electrochemical interfaces is the level set method, which tracks interface propagation over time based on the curvature-dependent speed.[13] This method has been used to model the roughness evolution during superconformal electrodeposition with additives and catalysts in the electrolyte.[14,15] With the level set method, the interface is considered to be sharp and is tracked and evolved based on its physical interactions, some of which relate to curvature. This method requires that the interface dynamics be understood a priori, making it difficult to extend to complex systems with many components.

Thermodynamically, the interface advances such that its free energy is reduced. One particular model for tracking an interface based on its free energy is the phase-field model. The phase-field approach uses one or more field variables to determine the phase of material at any location and time. The governing equation, derived from a thermodynamic statement of free energy or entropy, is coupled to the relevant transport equations of the system. Therefore, this model takes into account important physical phenomena during electrochemical interface shape evolution. It is also easily extended to complex multicomponent systems, as long as the thermodynamic free-energy function, surface energy, and diffusivity data are available. All phases are represented by one set of variables; thus, this method can easily handle a complex interface. Guyer et al. have developed a general formulation for phase-field modeling of electrodeposition and electrodissolution.[16,17] Although their approach includes the detail of charge distribution at the interface, it is currently limited in two ways: it requires discretization of the electrochemical double layer, limiting its use to nanometer-scale systems, and it is a numerically challenging technique, such that it has to date only been solved in one dimension.

Shibuta et al.[18] describe a simplified Allen–Cahn phase-field electrochemistry model to simulate a bridge formation in a nanometer-scale switch. Unfortunately, use of the Poisson equation with a dielectric constant as the governing equation for electrical potential is only correct when the charge at the interface is negligible, which in turn is only true for negligible current and interface resistance. It works reasonably when deposition is transport-limited and the ratio of electrode to electrolyte dielectric constant is qualitatively similar to the ratio of electrical conductivities, so their predicted shape evolution may be qualitatively correct. However, it is not clear that the space-charge effects can be neglected at such a small scale.

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Assadi[19] developed a Cahn–Hilliard phase-field model for electrodeoxidation in molten salt. This model used the continuity of current density as, the governing equation for electrical potential, except at the cathode-electrolyte interface. Its diffuse interface was between the metal and oxide in the cathode, and it assumed a sharp interface between the cathode and molten salt, with ohmic resistance at that sharp interface. The use of the sharp interface prevents its use to describe plating from the electrolyte, so this model is limited in application to electrodeoxidation; the assumption of ohmic resistance at the interface also limits it to small overpotential there.

When limited by mass transfer, the effect of charge-transfer resistance at the interfacial double layer is negligible, removing the need to discretize the double layer and permitting simulation of larger systems. Dussault and Powell[20-22] have reported two-dimensional results from the phase-field calculation of interfacial dynamics in an electrolytic steel refining process in which transport-limited reactions occur. This work has since been extended by Pongsaksawad and Powell.[23-25] In addition, the model can be coupled with fluid dynamics using an extension of Jacqmin's phase-field formulation of two-phase incompressible Navier–Stokes flows.[26] The simulation results can thus capture anode and cathode perturbation responses to an electric field for a solid or liquid system.

Building on the previous model of Dussault and Powell,[20-22] this paper presents two- and three-dimensional results for cathode shape and topology evolution in transport-limited electrolysis and rapid charge redistribution, including results for both all-solid and all-liquid binary (unsupported electrolyte) and ternary (supported electrolyte) systems, though mixed solid-liquid systems (e.g., solid electrode and liquid electrolyte) are a subject of ongoing study. The motivating applications are the electric-field-enhanced smelting and refining process (EFESR) in which iron is reduced from liquid slag containing iron oxide, and magnesiothermic reduction of titanium dichloride.[27] The conservation equations which comprise this model for a binary or ternary system, with and without fluid flow, are derived in the following section. The results of electrochemical interface evolution in the Fe–FeO system with and without convection are shown in the Model Results section; these results show that the model correctly reproduces the cathode interface stability criterion predicted by linear stability analysis.[2] That section also presents preliminary results from the ternary model for a transport-limited reaction in the Ti–Mg–Cl system. Results with convection (the all-liquid system) are then presented in the Discussion section. Finally, applications for this model and avenues of future expansion are discussed in the Conclusions.

Numerical Model

Phase-field formulation of electrochemistry.— The starting point for this work is a Cahn–Hilliard phase-field formulation for coupled fluid flow, diffusion, interfacial energy, electromigration, and electrochemical reactions developed by Dussault et al. and applied to a simple electrochemical system.[20] Cahn–Hilliard refers to a class of models in which a thermodynamic free-energy function of composition and its spatial derivatives determine the stable phases of a system and the energy and thickness of interfaces between them. Such models are also known as time-dependent Ginzburg–Landau and, among physicists, as "Model B."

Dussault's model did not include the electrostatic contribution to free energy. This study thus begins with the free-energy functional presented in the model of Guyer et al.,[16,17] which includes the electrostatic free energy $\Sigma z_i F c_i \Phi$ as well as terms for homogeneous chemical-free energy as a function of the dimensionless composition $\Psi(C)$ and gradient energy $|\nabla C|^2$

$$\mathcal{F} = \int \left[\beta \Psi(C) + \frac{\alpha}{2} |\nabla C|^2 + \sum z_i F c_i \Phi \right] dV \qquad [1]$$

where α and β are phase-field parameters which can be adjusted to control the thickness of the diffuse interface and interfacial energy. For a planar interface without electrical potential gradient, the interface thickness and energy scale are $\varepsilon \sim \sqrt{\alpha/\beta}$ and $\gamma \sim \sqrt{\alpha\beta}$, respectively. The electrostatic term includes the sum of electrostatic energy of each ionic species, where z_i represents the number of electron charges per ion, c_i is the molar concentration of each species, and Φ is the electrical potential.

For a two-phase system, the homogeneous free energy (Ψ) has two minima at the equilibrium compositions (or more generally, common tangents at those compositions; here we consider a case with two equal minima). A simplified homogeneous free-energy expression can be written as a polynomial function of dimensionless composition C

$$\Psi = C^2(1 - C)^2 \qquad [2]$$

The two minima are at $C = 0$, here representing the electrolyte, and $C = 1$, representing the metal; thus, the metal/electrolyte interface has C varying from 0 to 1. To write Eq. 1 in terms of C, the molar concentrations, c_i can be replaced by mole fractions

$$c_{\mathrm{Fe}} = \frac{\rho X'_{\mathrm{Fe}}}{X_{\mathrm{Fe}} M_{\mathrm{Fe}} + X_{\mathrm{O}} M_{\mathrm{O}}} \qquad [3]$$

$$c_{\mathrm{O}} = \frac{\rho X_{\mathrm{O}}}{X_{\mathrm{Fe}} M_{\mathrm{Fe}} + X_{\mathrm{O}} M_{\mathrm{O}}} \qquad [4]$$

where X_i is the mole fraction $X_{\mathrm{Fe}} = 0.5(1 + C)$ and $X_{\mathrm{O}} = 0.5(1 - C)$, ρ is the overall density, and M_i is the molar mass. For simplicity, total molar density is defined as $\rho/M = \rho/(X_{\mathrm{Fe}} M_{\mathrm{Fe}} + X_{\mathrm{O}} M_{\mathrm{O}})$ and assumed to be uniform in this model, as is approximately the case in the Fe–FeO system. Thus, the free-energy functional becomes

$$\mathcal{F} = \int \left[\beta \Psi(C) + \frac{\alpha}{2} |\nabla C|^2 + z_{\mathrm{Fe}} F \frac{\rho}{M} (1 + C)\Phi + z_{\mathrm{O}} F \frac{\rho}{M} (1 - C)\Phi \right] dV \qquad [5]$$

Here, we depart from Guyer's model and follow Dussault by assuming charge neutrality everywhere. This assumption restricts this model to mass transfer-limited electrochemical interactions with rapid charge redistribution; the assumption's validity under these restrictions is discussed in the Conservation of Charge section. However, this assumption greatly simplifies the model relative to that of Guyer, as it reduces the number of species needed to describe the system and removes the need to discretize the double layer. This simplification makes our model practically useful for systems on scales as large as millimeters, instead of tens of nanometers.

Assuming oxygen exhibits uniform charge ($z_{\mathrm{O}} = -2$) everywhere and is negligibly soluble in metallic iron, charge neutrality requires that the iron charge be given by

$$z_{\mathrm{Fe}}(C) = 2\frac{1 - C}{1 + C} \qquad [6]$$

Thus, the charge of iron is +2 in the electrolyte and zero in the metallic phase. Substituting $z_{\mathrm{Fe}}(C)$ into Eq. 5 eliminates the electromigration term. For this reason, to derive the electrochemical potential, constant charge in the bulk electrolyte is assumed for now, and then the variation of the iron charge is included later.

Thus, in the electrolyte, $z_{\mathrm{Fe}} = +2$ and $z_{\mathrm{O}} = -2$, the free-energy functional is reduced to

$$\mathcal{F} = \int \left[\beta \Psi(C) + \frac{\alpha}{2} |\nabla C|^2 + \frac{4F\rho}{M} \Phi C \right] dV \qquad [7]$$

The total electrochemical potential of iron is the variational derivative of the free-energy functional which is expressed as

$$\mu_{\mathrm{Fe}} = \frac{\delta \mathcal{F}}{\delta C} = -\alpha \nabla^2 C + \beta \Psi'(C) + \frac{4F\rho}{M} \Phi \qquad [8]$$

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Conservation equations: binary system.— Conservation of species.— The transport equation for conservation of iron in a fluid with velocity field $\vec{U}$ is

$$\frac{\partial C}{\partial t} + \vec{\nabla} \cdot (\vec{U}C) = - \nabla \cdot (\vec{J}_{\mathrm{Fe}}) \quad [9]$$

where $\vec{J}_{\mathrm{Fe}}$ is the flux in m/s. The variable iron charge depends on C across the interface; therefore, the flux can be derived by fitting the charge-neutrality condition as the electrolyte approaches the metal phase. Let μ'_{Fe} be the chemical potential of Fe without the electrostatic term $[\mu'_{\mathrm{Fe}} = -\alpha\nabla^2 C + \beta\Psi'(C)]$. Then, because iron electromigration flux is proportional to its charge z_{Fe}, that term can be multiplied by $(1 - C)/(1 + C)$ in order to make it applicable in both electrolyte and metal. Thus, the total flux includes a diffusion term driven by the gradient in chemical potential $\vec{\nabla}\mu'_{\mathrm{Fe}}$ and a migration term driven by the gradient in electrical potential $\vec{\nabla}\Phi$

$$\vec{J}_{\mathrm{Fe}} = -\kappa_{\mathrm{Fe}} \nabla \mu_{\mathrm{Fe}} = -\kappa_{\mathrm{Fe}}\left[\vec{\nabla}\mu'_{\mathrm{Fe}} + \frac{4F\rho}{M}\frac{(1 - C)}{(1 + C)}\vec{\nabla}\Phi\right] \quad [10]$$

where κ_{Fe} is the iron mobility. For iron and ferrous oxide, the molar density are roughly the same, so this term can be taken out of the gradient. Therefore, the complete conservation equation is given by

$$\frac{DC}{Dt} = \vec{\nabla} \cdot \left[\kappa_{\mathrm{Fe}}\vec{\nabla}\mu'_{\mathrm{Fe}} + 4\kappa_{\mathrm{Fe}}F\frac{\rho}{M}\frac{(1 - C)}{(1 + C)}\vec{\nabla}\Phi\right] \quad [11]$$

where D/Dt is the substantial derivative which represents the rate of change in the frame of reference moving with velocity $\vec{U}$. Mobility κ_{Fe} is assumed to be uniform and related to diffusivity D_{Fe} according to

$$D_{\mathrm{Fe}} = \frac{\kappa_{\mathrm{Fe}}\beta\left(\frac{d^2\Psi}{dC^2}\right)}{(c_{\mathrm{metal}} - c_{\mathrm{slag}})} \quad [12]$$

Equation 11 can be compared to that of Dussault's model,[20] which can be expressed in terms of

$$\frac{DC}{Dt} = \vec{\nabla} \cdot (\kappa_{\mathrm{Fe}}\vec{\nabla}\mu'_{\mathrm{Fe}}) + \vec{\nabla} \cdot \left[\frac{\sigma_{\mathrm{Fe}}(C)}{z_{\mathrm{Fe}}(C)F(\rho/M)}\vec{\nabla}\Phi\right] \quad [13]$$

such that σ_{Fe} is ionic conductivity defined as

$$\sigma_{\mathrm{Fe}}(C) = 4\kappa_{\mathrm{Fe}}F^2\left(\frac{\rho}{M}\right)^2\frac{(1 - C)}{(1 + C)} \quad [14]$$

Conservation of charge.— The equation for conservation charge can be expressed by

$$\frac{D\rho_f}{Dt} = -\vec{\nabla} \cdot [\vec{J}_{\mathrm{Fe}}Fz_{\mathrm{Fe}}(C) + \vec{J}_{\mathrm{O}}Fz_{\mathrm{O}} - \sigma_e(C)\vec{\nabla}\Phi] \quad [15]$$

where ρ_f is the charge density and electronic conductivity $\sigma_e(C)$ is estimated by linear interpolation between $\sigma_{e,\mathrm{metal}}$ and $\sigma_{e,\mathrm{slag}} \cong 0$. Then, from the definition of mole-weighted velocity, oxygen and iron fluxes are equal and opposite, $\vec{J}_{\mathrm{O}} = -\vec{J}_{\mathrm{Fe}}$. Thus, the conservation of charge equation reduces to

$$\frac{D\rho_f}{Dt} = -\vec{\nabla} \cdot \{\vec{J}_{\mathrm{Fe}}F[z_{\mathrm{Fe}}(C) - z_{\mathrm{O}}] - \sigma_e(C)\vec{\nabla}\Phi\} \quad [16]$$

Defining $\tilde{z}(C) = z_{\mathrm{Fe}}(C) - z_{\mathrm{O}}$ with $z_{\mathrm{Fe}}(C)$ given by Eq. 6 and substituting $\vec{J}_{\mathrm{Fe}}$ with Eq. 10, the above equation becomes

$$\frac{D\rho_f}{Dt} = \vec{\nabla} \cdot [\tilde{z}(C)F\kappa_{\mathrm{Fe}}\vec{\nabla}\mu'_{\mathrm{Fe}}] + \vec{\nabla} \cdot \left[\tilde{z}(C)4F\kappa_{\mathrm{Fe}}\frac{\rho}{M}\frac{(1 - C)}{(1 + C)}\vec{\nabla}\Phi\right] + \vec{\nabla} \cdot [\sigma_e(C)\vec{\nabla}\Phi] \quad [17]$$

To simplify the conservation of charge equation, we determine the (*i*) timescale of charge buildup relative to diffusion, (*ii*) current driven by diffusion relative to electromigration, and (*iii*) flux due to convection relative to current as follows.

Charge buildup vs diffusion timescale.— The timescale to form a steady interface total charge density is compared to the diffusion timescale. A one-dimensional diffusion model gives the conservation of charge as

$$\frac{\partial\rho_f}{\partial t} = \frac{\partial}{\partial x}\left(\sigma\frac{\partial\Phi}{\partial x}\right) \quad [18]$$

The charge density, ρ_f, divided by the permittivity, ϵ, yields Gauss's law

$$0 = \frac{\partial^2\Phi}{\partial x^2} + \frac{\rho_f}{\epsilon} \Rightarrow \rho_f = -\epsilon\frac{\partial^2\Phi}{\partial x^2} \quad [19]$$

By taking the derivative of Eq. 19 with respect to time and equating that with Eq. 18, the timescale to reach steady interface total charge density is estimated as follows

$$\frac{\partial\rho_f}{\partial t} = -\epsilon\frac{\partial}{\partial t}\frac{\partial^2\Phi}{\partial x^2} = \frac{\partial}{\partial x}\left(\sigma\frac{\partial\Phi}{\partial x}\right) \quad [20]$$

Therefore, $\tau_{\mathrm{charge}} \sim \epsilon/\sigma \sim 10^{-12}$ s.

The diffusion timescale can be estimated from the interface velocity and diffusion length

$$\tau_{\mathrm{diff}} = \frac{L_{\mathrm{diff}}}{U_{\mathrm{int}}} \quad [21]$$

where L_{diff} is the diffusion length or the boundary layer thickness and U_{int} is the interface velocity $[\tilde{z}(C)\sigma\Phi M/4FL_\rho]$. Here, the diffusion length is estimated by a 1D diffusion steady-state model as follows

$$D\frac{d^2c}{dx^2} + \left(\frac{zF}{RT}D\frac{d\Phi}{dx} + U\right) = 0 \quad [22]$$

which yields a diffusion lengthscale of

$$L_{\mathrm{diff}} \sim \frac{1}{\frac{zF}{RT}\frac{d\Phi}{dx} + \frac{U}{D}} \sim 0.11 \text{ mm} \quad [23]$$

Under the electrical potential of 1 V and the electrode separation of 0.5 cm, U_{int} is approximately 8.6×10^{-7} m/s. Thus, the diffusion time is around 130 s, which suggests that the timescale of interface motion due to diffusive processes is several orders of magnitude longer than the time required to establish steady-state charge accumulation at the interfaces. Therefore, the conservation of charge equation is effectively at steady state.

Diffusion vs electromigration.— The diffusion term is compared to the electromigration term by defining a dimensionless parameter, π_1

$$\pi_1 = \frac{\tilde{z}(C)F\vec{\nabla} \cdot (\kappa_{\mathrm{Fe}}\vec{\nabla}\mu'_{\mathrm{Fe}})}{\vec{\nabla} \cdot (\sigma\vec{\nabla}\Phi)} \quad [24]$$

In the binary model, the dimensionless concentration in the slag is zero and that in the metal is one. The chemical potential is thus uniformly equal to zero in the one-phase regions, as is its gradient. Because the system is assumed to be limited by mass transport away from the interfaces, net transport of charge due to chemical potential gradient can be neglected everywhere for this binary system. Thus, the first term on the right side of Eq. 17 can be eliminated.

Convection vs current.— The transport of charge by convection is compared to that by electrical field. A dimensionless term, π_2, is defined as

$$\pi_2 = \frac{\vec{U} \cdot \vec{\nabla}\rho_f}{\vec{\nabla} \cdot (\sigma\vec{\nabla}\Phi)} \sim \frac{U(\rho_f/\delta)}{\sigma(\Delta\Phi/L)/\delta} = \frac{U\epsilon}{L\sigma} \quad [25]$$

where U is the characteristic convective velocity (m/s) and δ is the Debye length. Even with high fluid velocity of 1 m/s and electrode

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separation of 0.5 cm, this dimensionless parameter is in the order of 10^{-10} ($\ll 1$). Therefore, convection can be safely neglected in the charge-transport equation.

The conditions described above are referred to as rapid charge redistribution, steady-state charge transport, and no convection. Finally, the conservation of charge equation (Eq. 17) becomes

$$0 = \vec{\nabla} \cdot \left[\tilde{z}(C) 4F\kappa_{\mathrm{Fe}} \frac{\rho}{M} \frac{(1-C)}{(1+C)} \vec{\nabla}\Phi \right] + \vec{\nabla} \cdot [\sigma_e(C)\vec{\nabla}\Phi] \quad [26]$$

where $\sigma_e(C)$ is expressed as $(3C^2 - 2C^3)\sigma_{e_0}$ and σ_{e_0} is the electronic conductivity of iron. This gives a value of σ_e of zero in the slag, σ_{e_0} in the metal, and in between in the diffuse interface. Next, we can define an effective electrical conductivity as

$$\sigma_{\mathrm{eff}} = \tilde{z}(C) 4F\kappa_{\mathrm{Fe}} \frac{\rho}{M} \frac{(1-C)}{(1+C)} + \sigma_e(C) \quad [27]$$

Thus, the conservation of charge equation is simplified to

$$0 = \vec{\nabla} \cdot [\sigma_{\mathrm{eff}}(C)\vec{\nabla}\Phi] \quad [28]$$

Note that the charge redistribution in an aqueous system is a diffusive process in which the simplification derived above may not be valid, particularly when the electric field fluctuates rapidly. In the ternary system, the concentration boundary layer is large and the transport of charge due to chemical potential gradient must be taken into account.

Dimensional analysis.— Conservation of species.— The conservation of species equation (Eq. 13) is normalized by defining $\tilde{t} = (t/\lambda^2/\bar{D}) = (t/\lambda^2/\kappa_{\mathrm{Fe}}\beta\langle\Psi''\rangle)$, where $\langle\Psi''\rangle$ is the characteristic value of Ψ'', $\tilde{\kappa}_{\mathrm{Fe}} = 1$, $\tilde{\mu} = (\mu/\beta)$, $\tilde{\nabla} = \lambda\nabla$, $\tilde{L} = (L/\lambda)$, $\tilde{\sigma}_{\mathrm{Fe}} = [\sigma_{\mathrm{Fe}}/4F^2\kappa_{\mathrm{Fe}}(\rho/M)^2]$, and $\tilde{\Phi} = (\Phi - \Phi_{\mathrm{cathode}}/\Phi_{\mathrm{anode}} - \Phi_{\mathrm{cathode}}) = (\Phi - \Phi_{\mathrm{cathode}}/\bar{\Phi})$. The dimensionless equation is expressed as

$$\frac{DC}{D\tilde{t}} = \frac{1}{\langle\Psi''\rangle}\tilde{\nabla} \cdot (\tilde{\nabla}\tilde{\mu}_{\mathrm{Fe}}) + Pe\frac{L}{\lambda}\tilde{\nabla} \cdot \left[\frac{\tilde{\sigma}_{\mathrm{Fe}}(C)}{z_{\mathrm{Fe}}(C)F} \tilde{\nabla}\tilde{\Phi} \right] \quad [29]$$

where

$$Pe = \frac{\bar{\sigma}_{\mathrm{Fe}}\bar{\Phi}\lambda}{zF(\rho/M)L\bar{D}} = \frac{\bar{\Phi}zF(\rho/M)\varepsilon\lambda}{2\sqrt{18}\cdot 3.1\gamma L} \quad [30]$$

The Peclet number here represents the advance of the interface due to metal reduction compared to the smoothing of the interface due to diffusion; thus, it can be expressed as $U_{\mathrm{int}}\lambda/D$, where U_{int} is the interface velocity due to plating.

Conservation of charge.— The charge-balance equation is normalized by defining $\tilde{\sigma}_{\mathrm{eff}} = (\sigma_{\mathrm{eff}}/\sigma_{\mathrm{slag}})$, where $\sigma_{\mathrm{slag}} = \tilde{z}_{\mathrm{slag}}4F\kappa_{\mathrm{Fe}}(\rho/M)$. The dimensionless equation becomes

$$0 = \tilde{\nabla} \cdot \left[\left(\frac{\tilde{z}(C)}{4} \frac{(1-C)}{(1+C)} + \tilde{\sigma}_e(C) \right) \tilde{\nabla}\tilde{\Phi} \right] \quad [31]$$

Navier–Stokes equations.— The Navier–Stokes equations describe conservation of mass and momentum in a fluid system. The dimensionless velocity-vorticity formulation using the same time and length scales as the ions and charge transport can be expressed as

$$\tilde{\nabla}^2\tilde{u} + \frac{\partial\tilde{\omega}}{\partial\tilde{y}} = 0$$

$$\tilde{\nabla}^2\tilde{v} - \frac{\partial\tilde{\omega}}{\partial\tilde{x}} = 0$$

$$\frac{\partial\tilde{\omega}}{\partial\tilde{t}} + \vec{u} \cdot \tilde{\nabla}\tilde{\omega} = Sc\tilde{\nabla}^2\tilde{\omega} - N\tilde{\nabla} \times (C\tilde{\nabla}\tilde{\mu}) \quad [32]$$

where $\tilde{u}$ and $\tilde{v}$ are dimensionless velocity components in the x and y directions, $\tilde{\omega}$ is the dimensionless vorticity, Sc is the Schmidt number $(\eta/\rho D)$, η is viscosity, and N is the force parameter $(\gamma/\rho u^2\varepsilon)$ equivalent to an inverse Weber number. The vorticity transport equation includes the curl of the surface tension forcing term $(-C\tilde{\nabla}\tilde{\mu})$ given by Jacqmin,[26] which describes driving force due to interfacial curvature. It is noted that simulations for liquid-liquid cases assume uniform viscosity.

Conservation equations: ternary system.— The binary model is next extended to a ternary system. This section briefly describes the formulation used. The motivating example used here is to study the rate of formation and structure of Ti produced via magnesiothermic reduction of $TiCl_2$. An electronically mediated reaction (EMR) through an electronic conducting medium plays a significant role in forming the structure; this system, therefore, can demonstrate the ability of this formulation to model EMR. This titanium reduction reaction is in turn central to the subhalide reduction process invented by Okabe and Takeda.[28]

Conservation of species.— Based on thermodynamics of the multicomponent system, the free-energy functional $\mathcal{F}$ as a function of phase parameters can be expressed as a volume integral of the sum of all free-energy terms

$$\mathcal{F} = \int \left[\Psi_0 + K_{22}|\nabla C_2|^2 + K_{33}|\nabla C_3|^2 + \sum_i^n \frac{z_i F\rho}{M} C_3\Phi \right] dV \quad [33]$$

Here, Ψ_0 is the homogeneous free energy, which is a function of local composition, K_{ii} are the gradient penalty coefficients (corresponding to α in the binary system), and C_i are composition parameters defined as follows. To assign the value for four phases (Mg, Ti, $MgCl_2$, and $TiCl_2$), we define C_2 as the fraction of Ti metal or chloride (vs Mg metal or chloride) and C_3 as the fraction of Ti or Mg metal (vs their chlorides). Thus, C_2 is zero for Mg and $MgCl_2$ and unity in Ti and $TiCl_2$, whereas C_3 is zero in chloride phases and unity in pure metal phases. The final term in Eq. 33, $(z_iF\rho/M)C_3V$, is the electrostatic free energy contributed by charges on ions. Because the oxidation state is only a function of C_3, this term exists only for that parameter.

An approximate homogeneous free-energy model is derived from the binary free-energy models. At $T = 1100$ K, it is assumed that magnesium chloride and titanium chloride form an ideal solution, though the metals themselves separate into Ti-rich and Mg-rich phases. The metal phases and the chlorides are assumed to be regular solutions with miscibility gaps, such that they separate into M and MCl_2 phases. A homogeneous free-energy function which leads to this behavior is

$$\begin{aligned}\Psi_0 = {} & C_2 ln C_2 + (1-C_2) ln(1-C_2) + \Omega_{12}C_2C_3(1-C_2) + C_3 ln C_3 \\ & + (1-C_3) ln(1-C_3) + \Omega_{13}C_3(1-C_3) \\ & - (C_2 - 0.5)(C_3 - 0.5)\end{aligned} \quad [34]$$

The first two terms are entropic terms in the Ti–Mg and $TiCl_2$–$MgCl_2$ systems. The third term produces the miscibility gap between metallic Ti and Mg. The forth and fifth terms are entropic terms in metal-chlorides systems. The sixth term produces the miscibility gap between metals and chlorides. The final term is the chemical driving force for the reaction of $TiCl_2 + Mg \rightarrow Ti + MgCl_2$. In our simulation, the interaction coefficient between metals, Ω_{12}, is 2.5, and that between metal chlorides, Ω_{13}, is 2.25.

As with the binary model, the chemical potential for each phase μ_i is derived by taking the variational derivative of the free-energy functional, again assuming constant ion charge in the electrolyte

$$\mu_2 = \frac{\partial\Psi_0}{\partial C_2} - K_{22}\nabla^2 C_2 \quad [35]$$

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$$\mu_3 = \frac{\partial \Psi_0}{\partial C_3} - K_{33}\nabla^2 C_3 + \frac{zF\rho}{M}\Phi = \mu_3' + \frac{zF\rho}{M}\Phi \qquad [36]$$

Similar to the binary model, μ_3' is the chemical potential of metal ions. The flux of C_2 and C_3 is driven by the gradient of these potentials (μ_2 and μ_3).

It is noted that electromigration only appears in the electrolyte; therefore, as with the binary case, the electromigration flux is proportional to the cation charge. The conservation equations of each parameter can be written in terms of mobility M_{ii} and potential μ_i as

$$\frac{\partial C_2}{\partial t} + \vec{U}\cdot\nabla C_2 = M_{22}\nabla^2\mu_2 \qquad [37]$$

$$\frac{\partial C_3}{\partial t} + \vec{U}\cdot\nabla C_3 = M_{33}\nabla^2\mu_3' + M_{33}\frac{zF\rho}{M}\nabla\cdot\left(\frac{1-C_3}{1+C_3}\nabla\Phi\right) \qquad [38]$$

The mobilities of species driven by their chemical potentials are assumed to be of the same magnitude ($M_{22} = M_{33}$). The mobilities of species driven by other chemical potential (M_{23} and M_{32}) are assumed to be zero. Mobility can be expressed in terms of diffusivity D_{ij} according to Fick's law of diffusivity as

$$M_{ij} = \frac{D_{ij}}{\dfrac{\partial^2\Psi}{\partial C_j^2}} \qquad [39]$$

Dimensionless parameters are defined as $\tilde{t} = (t/\lambda^2/D) = (t/\lambda^2/M_{33}\langle\Psi''\rangle)$, where $\langle\Psi''\rangle$ is the characteristic value of Ψ'', $\tilde{\mu} = (\mu/\langle\Psi''\rangle)$, $\tilde{\nabla} = \lambda\nabla$, and $\tilde{\Phi} = (\Phi\text{-}\Phi_{\text{cathode}}/\Phi_{\text{anode}}\text{-}\Phi_{\text{cathode}}) = (\Phi\text{-}\Phi_{\text{cathode}}/\bar{\Phi})$. Thus, the dimensionless equations are expressed as

$$\frac{DC_2}{D\tilde{t}} = \tilde{\nabla}^2\tilde{\mu}_2 \qquad [40]$$

$$\frac{DC_3}{D\tilde{t}} = \tilde{\nabla}^2\tilde{\mu}_3' + \mathcal{B}\tilde{\nabla}\cdot\left(\frac{1-C_3}{1+C_3}\tilde{\nabla}\tilde{\Phi}\right) \qquad [41]$$

The dimensionless parameter $\mathcal{B}$ is the ratio of the electrostatic free-energy density to the diffusive free-energy density, $(zF(\rho/M)\Delta\Phi/\langle\Psi''\rangle)$. In the simulation, $\mathcal{B}$ is scaled to one.

Conservation of charge.— The conservation of charge equation in the ternary case is

$$\frac{D\rho_f}{Dt} = -\nabla\cdot(J_{\text{Mg}}Fz_{\text{Mg}} + J_{\text{Ti}}Fz_{\text{Ti}} + J_{\text{Cl}}Fz_{\text{Cl}} - J_e) \qquad [42]$$

Here, $z_{\text{Mg}} = z_{\text{Ti}} = 2(1 - C_3/1 + C_3)$ and $J_{\text{Mg}} + J_{\text{Ti}} = (J_{\text{Cl}}/2) = J_3$. The charge conservation equation is reduced to

$$0 = \tilde{\nabla}\cdot[\tilde{z}(C_3)FK_{33}\tilde{\nabla}\mu_3] + \tilde{\nabla}\cdot(\sigma_{\text{eff}}\tilde{\nabla}\tilde{\Phi}) \qquad [43]$$

where $\tilde{z}(C_3) = 2(1 - C_3/1 + C_3) + 2$ and $\sigma_{\text{eff}} = [\tilde{z}(C_3)/4][(1 - C_3)/(1 + C_3)] + [\sigma_e/\sigma_{\text{electrolyte}}]$. In this present study, we define σ_e as $\sigma_{e0}(3C_3^2 - 2C_3^3)(1 - (1/2)C_2)$. Thus, σ_e is zero in the chloride phases, whereas σ_e of Ti is half of that of Mg.

Model Results

The 2D and 3D formulations were discretized using the finite-difference method. Time derivatives were modeled using the Crank–Nicholson semi-implicit timestepping algorithm. The implementation was written using the open-source PETSc suite[29] from Argonne National Laboratories, which handled distributed storage of the discretized field variables, vectors and matrices, and distributed solution of the resulting system of nonlinear equations.

Binary Fe–FeO system.— The EFESR process[27] is the motivating application for the electrochemistry formulation developed in this study. This process operating in a steelmaking vessel accelerates the reaction between iron oxide in the slag and carbon in the steel by applying an electric field across the slag layer. The products of this process are pure iron at the cathode in the slag, purer iron metal forming the anode below, and CO bubbles at the anode/electrolyte interface.

Table I. Numeric parameters used in the binary Fe–FeO model in comparison to their physical values.

Parameters	Physical	Numeric
$\frac{\rho}{M}$	1.31×10^5 mol/m^3	1.31×10^5 mol/m^3
σ	12.0 Ω^{-1} m^{-1}	12.0 Ω^{-1} m^{-1}
σ_e	10^5 Ω^{-1} m^{-1}	10^3 Ω^{-1} m^{-1}

Here a simplified three-layer structure of the binary system (Fe–FeO–Fe) is used as an initial condition. Because the scale of the cathode instability is around 0.1 mm, the domain size is 0.1 $\times$ 0.2 mm. In the iron oxide slag C is set to zero, and in the iron metal C is set to one, with little fluctuation ($\sim$0.001) throughout the domain. The voltage at the upper boundary is set to 0 (cathode) and that at the lower layer is set to 1 (anode). Periodic boundary conditions are applied in the x direction. Table I displays physical and numerical parameters used in the binary system simulations. The value of σ_{e0} in the simulations is smaller than the actual value to improve convergence, while it is still relatively much greater than that of the slag. Model results are shown presently (Fig. 4c) to asymptotically approach the limit of perfect electrode conductivity for electronic to ionic conductivity ratio at or above 83, which is the value used here.

Two-dimensional model.— The interface instability due to change in electrical potential and surface energy is investigated first for the solid Fe–FeO system. One small box perturbation is introduced to each electrode interface (top: cathode and bottom: anode). The perturbation on the anode disappears because it is preferentially oxidized and dissolved due to the concentration of the electric field there. At a low surface energy shown in Fig. 1a, large electrical potential results in the formation of dendrites at the cathode, which may short-circuit the system or break into droplets, as observed in experiments. At a larger surface energy shown in Fig. 1b, the perturbation on the cathode is flattened.

Our phase-field model can capture the interface in more detail compared to the preliminary result of two-dimensional evolution of emerging dendrite in electrodeposition simulated by the electrochemical mean-field kinetic lattice model.[30] This model employed unrealistic simulation parameters due to lengthscale limitation. In the real systems, the lengthscale of the aggregation is in an order of micrometer, but the lattice model lengthscale is in Angstrom, Å. However, our model neglects the charge double-layer effect and allows us to simulate at much larger scales up to millimeters.

In the liquid Fe–FeO system, interface stability is also influenced by the fluid viscosity. In the limit of high viscosity, the system behaves like the solid system. But, as the fluid viscosity is reduced, flow provides a second mechanism to stabilize the interface as shown in Fig. 2.

These simulations are run at low resolution for two reasons. First, visualizations of the composition field more clearly show the diffuse interface when it is thick relative to the overall domain, and a thick interface does not require high resolution. Second, for the stability analysis benchmark simulations of the Stability Analysis for the Binary Model section, demonstrating reasonable accuracy for a simple sinusoidal perturbation at low resolution and with a thick interface permits confident application of the model to much more complex systems without requiring an extremely refined mesh.

Three-dimensional model.— A 3D binary model without flow under an unstable condition is carried out. The initial perturbation is a cubic shape on the electrode surfaces similar to those in 2D cases.

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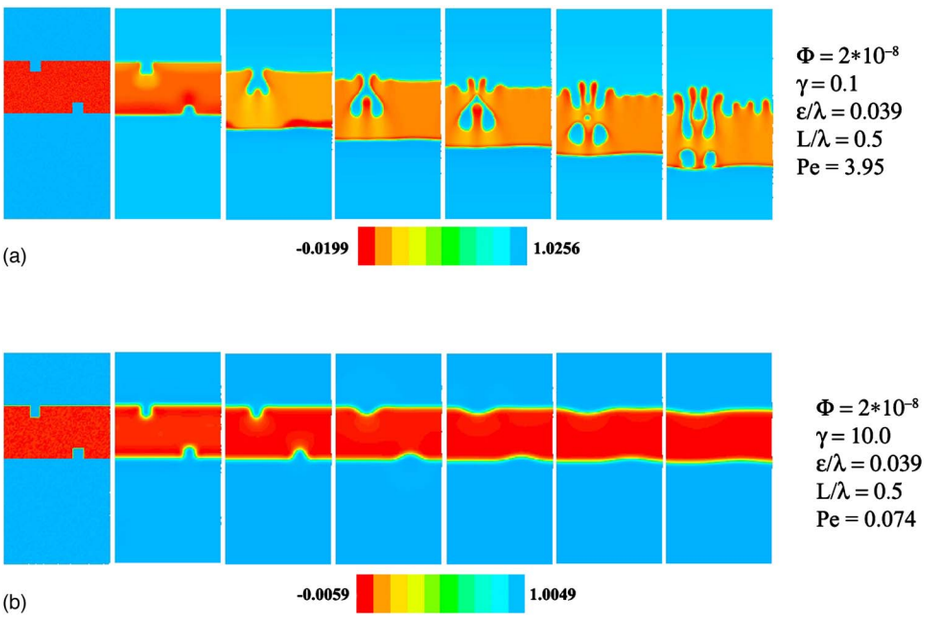


Figure 1. (Color online) Interface dynamics without convection: (a) cathode shape evolution under high electric field on a 150 × 300 grid (the perturbed cathode interface is unstable) and (b)cathode shape evolution under high surface tension on an 80 × 160 grid (the perturbed cathode interface is stable). The left contour represents the initial condition with a three-layer structure, metal/electrolyte/metal. The cathode interface evolves with time as shown in the figures on the right. The scales on the bottom of each figure indicate the average maximum and minimum compositions throughout each simulation domain.

The perturbation on the cathode grows into a long streamer which causes the roughness on the surrounding surface. The 3D diffuse interfaces are shown in Fig. 3. When the streamer arm reaches the anode, short circuit occurs. Finally, the whole cathode interface appears to be highly unstable.

Stability analysis for the binary model.— To verify our model, we perform a series of simulations with a sinusoidal perturbation on the cathode at varying applied potential and interfacial energy. The growth rate of the cathode roughness is estimated from the change in amplitude with time. Negative growth rate indicates stability. Positive growth rate shows instability. The critical magnitude of the electric field and interfacial energy at the dimensionless scales are recorded. Figure 4a shows the dimensional plot of the electric field as a function of the interfacial tension for an Fe–FeO solid-solid system. Linear plots from the phase-field simulations, labeled as B–D, are compared to the analytical result from Barkey's stability theory of a sharp interface,[2] labeled as A. The results suggest that the cathode interface is more stable in the low electric field and high interfacial energy region. The deviation from the sharp interface model is slightly larger as the separation between electrodes (L/λ) decreases and much larger as the electronic conductivity (σ_e/σ) decreases. Differences between this model and Barkey's results are discussed in the Validation of Model Results section.

The stability criterion can also be expressed in terms of dimensionless parameter *Pe* (see the Dimensional Analysis section), for large values of which the interface is unstable. Expressed this way, Barkey's model, which assumes a sharp interface, infinite electrode separation, and perfect electrode conductivity, can be reduced to a single critical value of the Peclet (*Pe*) number

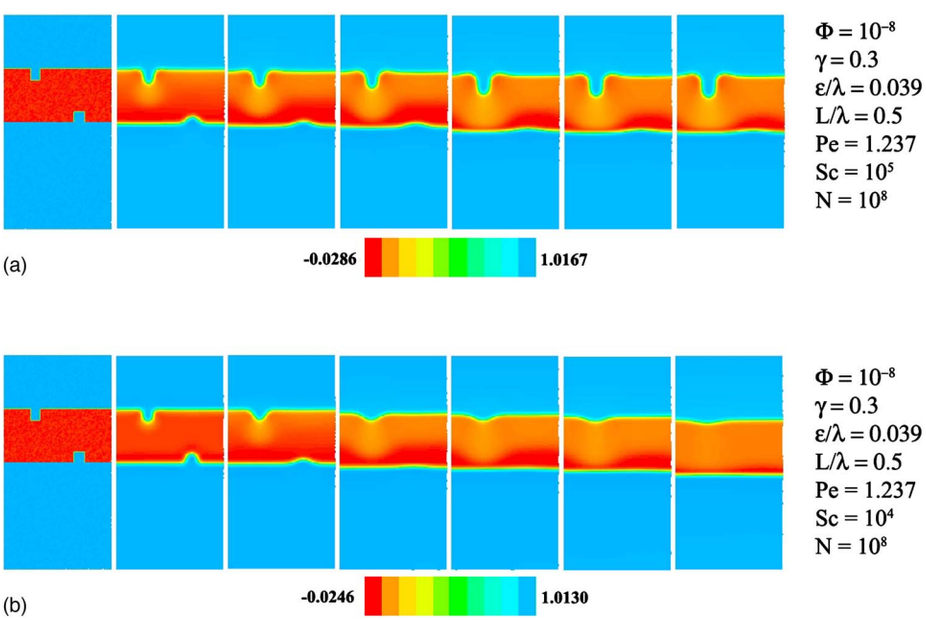


Figure 2. (Color online) Interface topologies with fluid flow on a domain size of 80 × 160. The left contour represents the initial condition with a three-layer structure, metal/electrolyte/metal. The cathode interface evolves with time as shown in the figures on the right. The scales on the bottom of each figure indicate the average maximum and minimum compositions throughout each simulation domain. (a) Cathode shape evolution under high viscosity fluid. The perturbed cathode interface is unstable. (b) Cathode shape evolution under low viscosity fluid. The perturbed cathode interface is stable.

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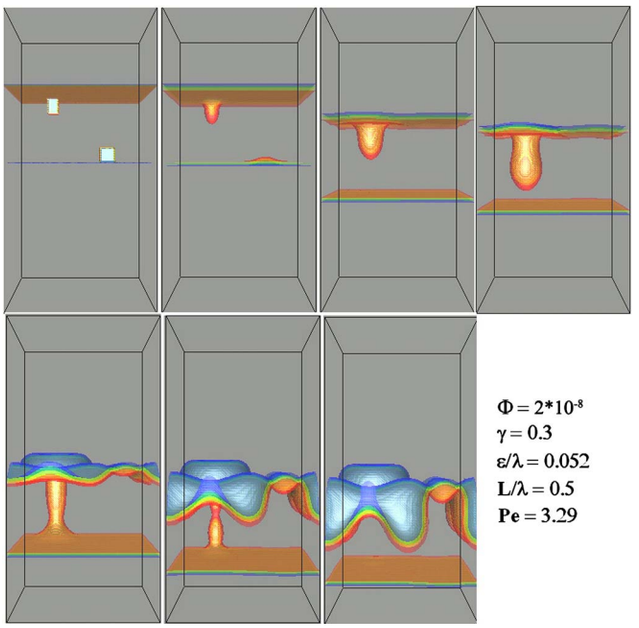


Figure 3. (Color online) 3D results of the cathode topology under electric field with no convection on the domain size of 60 × 120 × 60. The diffuse interfaces are represented by the composition contour surfaces: red, yellow, green, and blue at C = 0.2, 0.4, 0.6, and 0.8, respectively. The bump on the cathode grows and short-circuits when it reaches the anode.

$$Pe\frac{\lambda}{\varepsilon} = \frac{\bar{\Phi} z F(\rho/M)\varepsilon\lambda}{2\sqrt{18}\cdot 3.1\gamma L} = 0.116 \qquad [44]$$

To make this independent of interface thickness, we multiply by λ/ε

$$Pe\frac{\lambda}{\varepsilon} = \frac{\bar{\Phi} z F(\rho/M)\lambda^2}{2\sqrt{18}\cdot 3.1\gamma L} = 3.00 \qquad [45]$$

For a system without fluid flow, other dimensionless parameters in this model are L/λ, σ_e/σ, and ε/λ. To study the effect of different parameters on the interface instability, the critical value of Pe is plotted as a function of these parameters as shown in Fig. 4.

The second plot shown in Fig. 4b illustrates the change in the stability criterion as a function of the distance between the two electrodes under constant electric field. The result from the phase-field model shows less stability than Barkey's result, likely because the conductivity of the nearby anode enhances the higher density of field lines at the tip of the cathode perturbation.

Figure 4c shows the interface stability criterion for different values of the dimensionless electronic conductivity of the metal. It is observed that at high electronic conductivity, the interface is unstable and approaches the Barkey result of perfect electrode conductivity, then surpasses it due to finite electrode separation as explained previously. And, as the electronic conductivity of the metal approaches that of the slag (σ_e/σ = 1), Pe approaches infinity where the interface is always stable because the electric field is uniform and interfacial tension always stabilizes the perturbation.

From the results in Fig. 4d, the interface stability criterion for large σ_e/σ remains almost unchanged at various diffuse interface thicknesses. Referring to the Phase-Field Formulation of Electrochemistry section, the interface thickness scales approximately as $\varepsilon \sim \sqrt{\alpha/\beta}$ and the interfacial energy as $\gamma \sim \sqrt{\alpha\beta}$.[d] Decreasing β and increasing α while keeping the interfacial energy the same as the physical value exaggerates the interface thickness for simulation purposes. This plot confirms that the numerical diffuse interface thickness ranging from 0.039λ to 0.116λ in this phase-field model

[d] Under some circumstances with thick interfaces, it is observed that the cathode-electrolyte interface thickens and the anode-electrolyte interface thins; this effect was not significant in the simulations presented here, so this scaling behavior is valid.

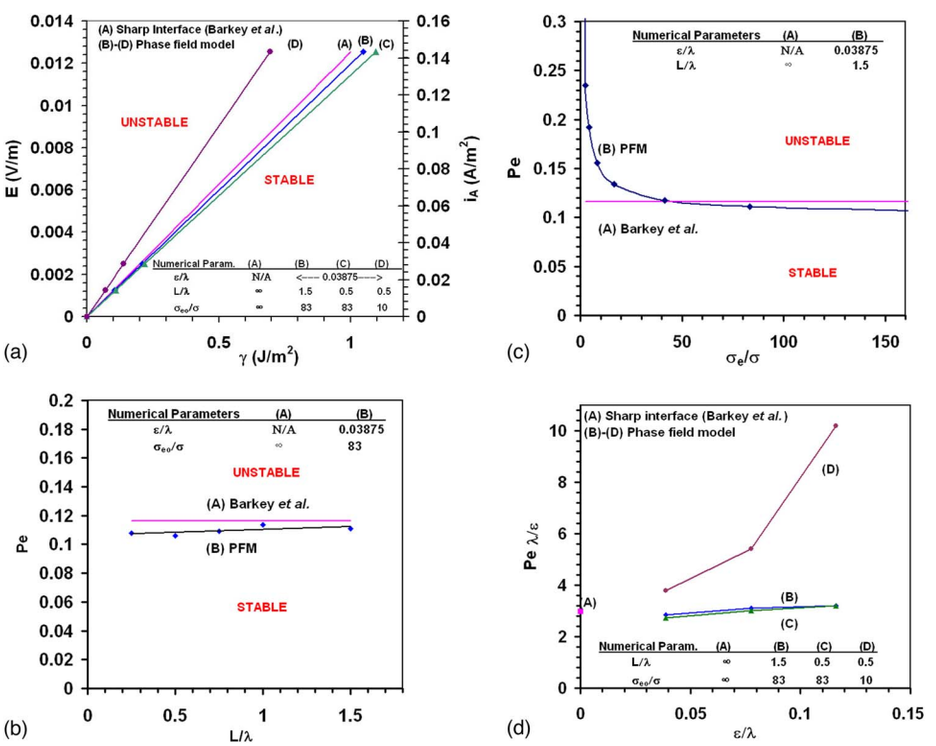


Figure 4. (Color online) Stability criteria with varying dimensionless parameters: (a) comparison with linear stability analysis, (b) varying separation between electrodes, (c) varying electronic conductivities, and (d) varying interface thickness.

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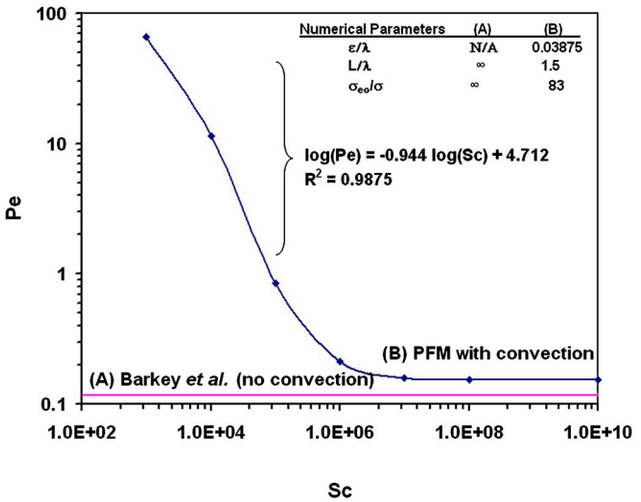


Figure 5. (Color online) Log *Pe* vs log *Sc*. The low *Sc* regime ($Sc < 10^6$) has the slope of −0.944, showing that the Peclet number is inversely proportional to the Schmidt number as expected.

can represent the physical diffuse interface thickness because the stability behavior is not affected, at least for large σ_e/σ shown by curve B and C.

However, curve D in that figure shows that for small σ_e/σ, the system is more stable with a thicker interface. In this case, the field lines are not normal to the interface as they would be for large σ_e/σ, making the interface more stable, as was shown in Fig. 4c. It makes intuitive sense that the diffuse interface enhances the deviation from normality, making the system more stable for a thicker interface, as is shown by curve D of Fig. 4d. For this reason, using this model to describe a system with low σ_e/σ requires a thinner interface to approach the sharp interface limit.

For the case with convection, the interface stability also depends on the Schmidt number *Sc*. The result shown in Fig. 5 indicates that less viscous fluid (low *Sc*) results in stronger flow and stabilizes the interface, whereas highly viscous flow (high *Sc*) does not. At low *Sc* on a log scale plot, the relationship between *Pe* and *Sc* is linear with the slope of −0.944, which agrees well with the stability analysis described in Appendix B.

At $T = 1773$ K, liquid FeO has the following properties: viscosity $\eta_{FeO} = 0.20$ Pa s,[31] diffusivity $D_{FeO} = 4.20 \times 10^{-7}$ m^2/s,[32] density $\rho_{FeO} = 3.50 \times 10^3$ kg/m^3, and interfacial tension $\gamma_{FeO} = 6.45 \times 10^{-1}$ N/m.[33] The value of *Sc* is 1.40×10^2. The extrapolated critical value of Pe_{liq} is therefore approximately 4.86×10^2, which is larger compared to *Pe* of solid ($Pe_{solid} = 0.116$). In the experiment of EFESR, a voltage of 2 V was applied across the electrolyte separated by 10 cm. The diameter of a quenched cathode streamer was 0.2 mm,[27] which corresponds to the primary dendrite spacing around 0.4 mm. With an applied electric field of 20 V/m and Eq. 30, the corresponding critical wavelength for the liquid-liquid system is approximately 0.6 mm. The simulation result is in the same order of magnitude compared to the experimental observation. The difference is due to the assumption that viscosity is uniform. In fact, the viscosity of FeO is greater than that of Fe. Future work of the binary model will take into account the variation of fluid properties.

Preliminary results of the ternary model (Ti–Mg–Cl).— Three sets of simulations were carried out to test the consistency and qualitative correctness of our ternary model. These are a one-dimensional simulation of the open circuit, a two-dimensional simulation of the closed circuit with no contact between Mg and $TiCl_2$, and a two-dimensional simulation with contact between those phases.

Table II. Numerical parameters used in the ternary system.

Parameter	Numeric value
$K_{22} = K_{33}$	5×10^{-4}
$M_{22} = M_{33}$	2×10^{-2}
σ	200 Ω^{-1} m^{-1}
σ_{e0}	100 Ω^{-1} m^{-1}

The initial conditions (C_2, C_3) were set as follows: Mg(0.02, 0.98), $MgCl_2$(0.02,0.02), $TiCl_2$(0.98,0.02), and Ti (0.98, 0.98). The electrical potential was set to one at the lower left pixel and zero elsewhere. The model parameters are shown in Table II. These values are chosen to promote fast convergence to the solution in each timestep.

One-dimensional model.— A one-dimensional model is used to assess the open-circuit potential of the $Mg/MgCl_2 \parallel TiCl_2/Ti$ system. The simulation domain is 5×100. The configuration is an open-circuit structure where Mg is on the left and Ti is on the right separated by $MgCl_2$ and $TiCl_2$, shown in Fig. 6a. The reaction proceeds by the diffusion of reactant ions with the following half-cell reactions

$$Mg \rightarrow Mg^{2+} + 2Cl^-$$

$$Ti^{2+} + 2e^- \rightarrow Ti$$

Accumulation of electrons at the $Mg/MgCl_2$ interface and positive charges at the $TiCl_2/Ti$ interface causes an electrical potential gradient across the chloride solution. As a result, this gradient eventually stops the formation of Ti. The concentration and electrical potential fields and gradient plots are shown in Fig. 6.

Two-dimensional model.— For the 2D model, the simulation runs on a 50×50 grid. Connecting Mg and Ti closes the circuit, allowing the electrons from the $Mg/MgCl_2$ interface to be conducted through these metal phases to the $TiCl_2/Ti$ interface and permitting more Ti deposition. This process is known as an electronically mediated reaction (EMR). This phase-field simulation proceeds in a qualitatively correct way as shown in Fig. 7. The evidence of EMR is the voltage gradient across the metal phases; voltage is high where positive charge is produced at the $TiCl_2/Ti$ interface and low where electrons are produced at the $Mg/MgCl_2$ interface, as expected. Thus, electrons move from low to high electrical potential (because of their negative charge), but the chloride ions move from high to low electrical potential because of the chemical driving force, allowing the reaction to proceed toward completion.

To further demonstrate the EMR through the product phase, Mg and $TiCl_2$ are set up to be in direct contact (see Fig. 8). The titanium phase formed at that interface in contact with the dissolving magnesium serves as an electronic conducting path for EMR. The voltage gradient across the Ti product confirms the Ti formation by EMR.

Discussion

Validation of model results.— The binary (unsupported electrolyte) simulations demonstrate that the interface instability is affected by the electric field in the electrolyte, surface energy, material properties, and in the liquid-liquid case, viscosity. The results without fluid flow effect are in good agreement with the published analytical stability criterion (see Appendix A). The deviation from the analytical result (Fig. 4) seems to arise from three differences between the analytical and numerical models. The first condition is the finite separation between electrodes, which reduces stability. The second condition is the finite cathode conductivity assumed in the phase-field model, which enhances stability. Finally, the last condition is the diffuse interface introduced by the phase-field model, for which results show that a more diffuse interface slightly enhances stability for large conductivity ratio and greatly enhances it for small conductivity ratio.

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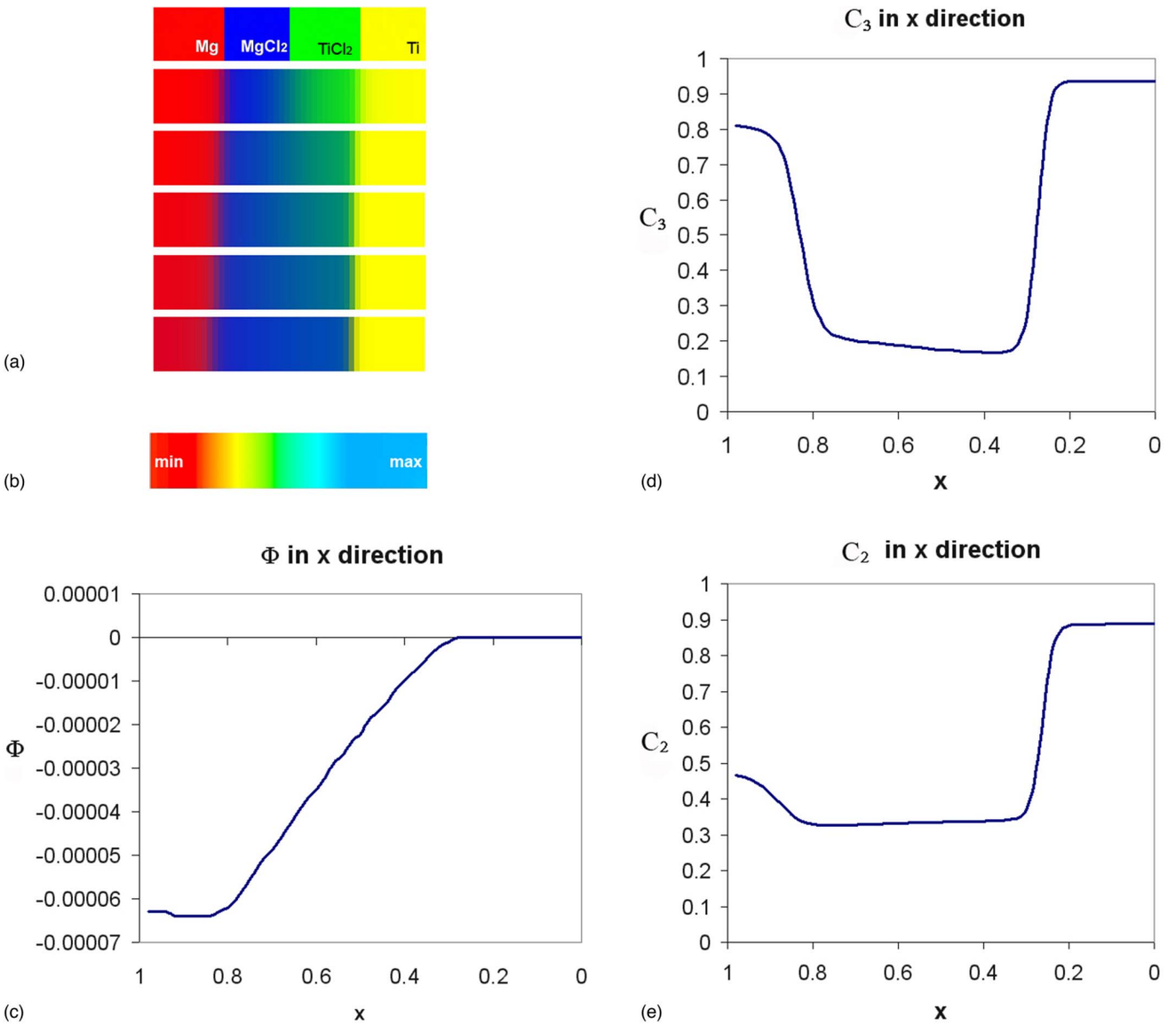


Figure 6. (Color online) Long-time solution in the 1D open-circuit model. The C_2 and C_3 values show magnesium, interdiffused chlorides, and titanium phases appearing from left to right. A voltage gradient is caused by electron accumulation at the $Mg/MgCl_2$ interface and electron depletion at the $TiCl_2/Ti$ interface. (a) Concentration contours as time increases (top to bottom). (b) Voltage contour at the final time step. (c-e) Plots of electrical potential and composition parameters at the end of the simulation.

The results of simulations with fluid flow are validated with the stability analysis based on a scaling solution derived in Appendix B. At low Schmidt number (*Sc*), the effect of flow is significant, and the results are in excellent agreement with the stability analysis. At high *Sc*, the interface shape and topology evolution approach solid behavior.

The ternary (supported electrolyte) simulations agree qualitatively with predictions based on experimental evidence of reduction of titanium via EMR.[34] The simulations demonstrate that the magnesiothermic reduction of titanium by EMR occurs when there exists an electronically conducting medium for transporting electrons. Quantitative validation of model results based on the stability analysis in an arbitrary supporting electrolyte system (by Haataja et al.[9]) is currently under investigation.

Limitations and model applications.— The electrochemical reaction in this model is in the mass-transport-limited régime. Under this condition, the mass-transfer resistance is much greater than the charge-transfer resistance. As a result, cations recombine with electrons as soon as the ions arrive at the cathode interface. At high temperatures such as in metal smelting processes, the mass-transport-limited condition is valid. However low-temperature processes such as an aqueous electrolysis system are in the charge-transfer-limited or mixed-limitation regimes. Accumulation of cation in front of the cathode leads to a voltage drop across the interface, which is not captured by this model.

One approach to capturing charge-transfer resistance would make the conductivity very small for intermediate values of C, creating an ohmic resistance at the interface. As discussed by Assadi,[19] this would extend the applicability of this model from negligible resistance to the regime of small overpotential for which the interface resistance can be considered ohmic. However, whereas Assadi's model straightforwardly incorporated that resistance at a sharp interface, accurately representing it at a diffuse interface would require very uniform thickness; it is observed that the cathode-

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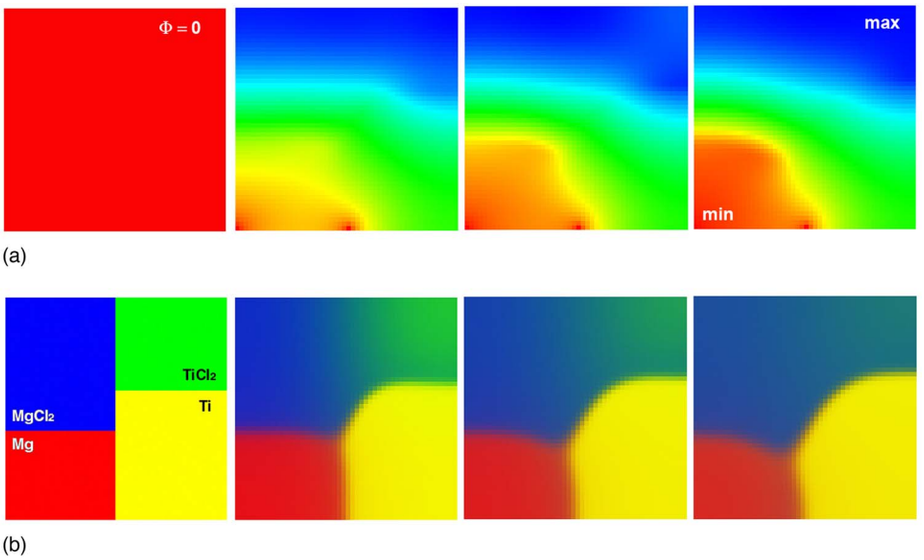


Figure 7. (Color online) Closed-circuit simulation with Ti and Mg phases in contact. In the electrical potential contour plot, red indicates low and blue indicates high value of electrical potential, as labeled by min and max on the figure. The Mg, $MgCl_2$, $TiCl_2$, and Ti phases are colored red, blue, green, and yellow, respectively, as also shown by labels. Ti formation is by chemical reduction at the Ti–Mg–chlorides triple point and by EMR between the Ti–$TiCl_2$ and Mg–$MgCl_2$ interfaces. The resolution is 50×50. (a) Electrical potential field. (b) Phase field.

electrolyte interface thickens slightly and the anode-electrolyte interface thins with a plating current. Furthermore, accurately representing the variation of conductivity across the thin diffuse interface would require a very fine grid there.

Moreover, the transport of chemical species is assumed to be independent of temperature. Scaling analysis is required to confirm that the thermal gradient has no significant effect on mass and charge transports for the isothermal model here to be valid. Alternatively, a heat conduction model could be coupled with this model to capture this effect with high electrical current and high electrical resistivity. Joule heating needs to be taken into account as another heat source for such a model.

In addition to the EFESR process for iron, this model can be applied to magnesium or aluminum reduction, both of which are two-phase liquid-liquid systems with transport-limited electrochemical reactions. This phase-field model can serve as a tool to study cathode shape and topology evolution. For magnesium, this would assist in development of an efficient cathode design; for aluminum, it could help to determine conditions for suppression of harmful streamers. Some of these systems exhibit significant electrocapillarity effects where interfacial energy is a function of the perpendicular component of electric field. This phenomenon can be captured by a straightforward extension of the present model, by making the interfacial energy (determined by free energy of mixing β and gradient penalty coefficient α in the binary case) depend on the dot product of electric field and composition gradient

$$\gamma = f(\nabla\Phi \cdot \nabla C) \text{ or } f(\nabla\Phi \cdot \nabla C_3) \quad [46]$$

This can be captured without modeling the behavior of the double layer in detail.

Further modification to this model is necessary to include multiphase (solid/liquid/gas) systems and those with nonuniform mobility and fluid properties. A model of mixed solid/liquid systems with stationary solids is based on the formulation developed by Tonhardt and Amberg[35] and Beckermann et al.[36,37] in the study of dendritic growth in a shear flow. A model with moving solids is based on the mixed stress formation of Powell and Dussault.[38] Solids also introduce complexity associated with anisotropy, leading to crystal faceting or proffered directions for dendrite growth. Modeling such systems requires keeping track of crystal orientation, such as the models of Warren et al.[39] A model incorporating these effects would

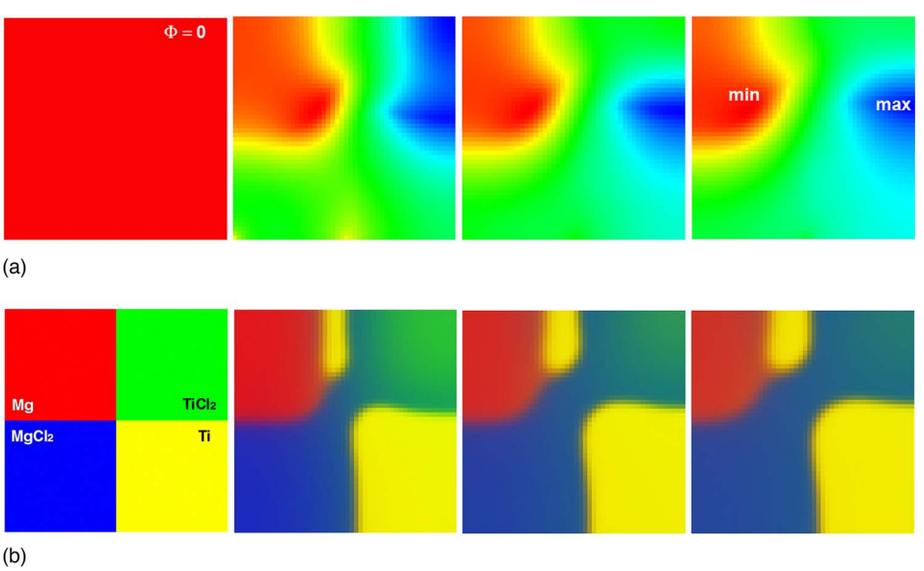


Figure 8. (Color online) 2D model with $Mg/TiCl_2$ in contact. Composition and color scales are the same as those of Fig. 7. Titanium is first formed by a chemical reaction at the $Mg/TiCl_2$ interface. More titanium plating then proceeds by EMR as indicated by the change in voltage across that titanium phase. The resolution is 50×50. (a) Voltage field. (b) Phase field.

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be a powerful tool for efficiently predicting polycrystalline microstructure produced by transport-limited electrolysis processes.

Conclusions

Electrochemical interface instability leads to surface roughness and dendrites in several processes from metal smelting to electrodeposition. In this work, the phase-field method is used to understand and predict this behavior. For a binary model, our work presents two- and three-dimensional results showing the instability on the cathode interface in situations of transport-limited electrolysis, such as the EFESR process (Fe–FeO system).

The two-dimensional model has been verified by comparing sinusoidal perturbation simulations with the linear stability theory. In general, the interface is unstable under high voltage, low surface tension, and high viscosity as expected. Three-dimensional results illustrate the interface in greater detail and correlate well with the two-dimensional cases.

Furthermore, the model has been extended to a ternary system in which the mechanisms of magnesiothermic reduction of titanium are investigated. The preliminary results in one and two dimensions are in good qualitative agreement with the theory. The model is easily extended to more complex multicomponent systems with electrocapillarity or solid-liquid interaction. The interfacial double layer need not be modeled to use this approach for many applications satisfying the assumptions of transport-limited operation and rapid charge redistribution.

Acknowledgment

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Appendix A

Simplified stability analysis (without fluid flow).— According to Barkey et al.[2] stability of a sinusoidal perturbation on electrochemical interface depends on the electric field, the fraction of limiting current, and the surface energy. The interface is stable when the sum of electric field and the fraction of limiting current is less than the surface energy term

$$\frac{i_A}{\sigma} + \frac{RT\dfrac{i}{i_L}}{\delta n F\left(1 - \dfrac{i}{i_L}\right)} < \frac{2\gamma\left(\dfrac{2\pi}{\lambda}\right)^2}{zF(\rho/M)} \qquad [A\text{-}1]$$

where i_A is the current density for a flat interface, R is the gas constant, T is the operating temperature, i_L is the diffusion-limiting current density, δ is the Nernst diffusion layer thickness, and λ is the wavelength of the sinusoidal perturbation. The first term represents the growth rate due to an electric field. The second term represents the growth due to diffusion in the boundary layer which does not exist in an unsupported ionic electrolyte. The last term is the flattening of the interface due to surface tension. For this study with an unsupported ionic electrolyte, the stability criterion is reduced to

$$\frac{i_A}{\sigma} < \frac{2\gamma\left(\dfrac{2\pi}{\lambda}\right)^2}{zF(\rho/M)} \qquad [A\text{-}2]$$

Rearranging this equation by replacing the current/conductivity ratio in the electric field term with the potential gradient yields

$$\frac{\Delta\Phi}{L} < \frac{8\pi^2\gamma}{zF\lambda^2(\rho/M)} \qquad [A\text{-}3]$$

The potential can be plotted as a function of surface energy as

$$\Delta\Phi = \frac{8\pi^2 L\gamma}{zF\lambda^2(\rho/M)} \qquad [A\text{-}4]$$

The slope of this plot is compared to that from the simulation results as shown in Fig. 4a.

Appendix B

Stability analysis (with fluid flow).— The growth rate of a small amplitude (A) perturbation is derived by Barkey et al.[2] In the mass-transfer-limited regime, the growth rate due to electric field and surface energy is expressed by

$$\frac{1}{A}\frac{\partial A}{\partial t} = \frac{\sigma\left(\dfrac{2\pi}{\lambda}\right)}{zF(\rho/M)} \frac{\left(\dfrac{i_A}{\sigma} - \dfrac{2\gamma M\left(\dfrac{2\pi}{\lambda}\right)^2}{zF\rho}\right)}{\dfrac{1}{\sqrt{2}} + \dfrac{RT}{azFi_A}\sigma\left(\dfrac{2\pi}{\lambda}\right)} \qquad [B\text{-}1]$$

where a is equal to 0.25. To incorporate the effect of viscosity, consider a critically stable perturbed liquid-liquid cathode-electrolyte interface. In the frame of reference of that interface, the momentum equation in velocity-pressure form scales as follows

$$0 = -\nabla P + \eta\nabla^2\vec{U} \sim \frac{\Delta P}{\lambda} + \frac{\eta u}{\lambda^2} \qquad [B\text{-}2]$$

where P is pressure which scales as the second derivative of curvature $\Delta P \sim (\gamma A/\lambda^2)$. Substituting this for P and solving for u(m/s) leads to a velocity scale given by

$$u \sim \frac{\gamma A}{\lambda\eta} \qquad [B\text{-}3]$$

This scaled velocity is equal to the rate at which the perturbation amplitude would shrink in the absence of other phenomena ($-\partial A/\partial t$). In the limit of a low Schmidt number (Sc), the flow has a larger effect on the interface stability than the diffusive term (second term in the right-side numerator of Eq. B-1), which can therefore be neglected. Adding this flow term to amplitude growth rate gives

$$\frac{1}{A}\frac{\partial A}{\partial t} = \frac{\sigma\left(\dfrac{2\pi}{\lambda}\right)}{zF(\rho/M)} \frac{\left(\dfrac{i_A}{\sigma}\right)}{\dfrac{1}{\sqrt{2}} + \dfrac{RT}{azFi_A}\sigma\left(\dfrac{2\pi}{\lambda}\right)} - \frac{k_f\gamma}{\lambda\eta} \qquad [B\text{-}4]$$

where k_f is a flow scaling constant. When the right side is negative, the interface is stable. Therefore, the stability criterion becomes

$$\frac{\sigma\left(\dfrac{2\pi}{\lambda}\right)}{zF(\rho/M)} \frac{\left(\dfrac{i_A}{\sigma}\right)}{\dfrac{1}{\sqrt{2}} + \dfrac{RT}{azFi_A}\sigma\left(\dfrac{2\pi}{\lambda}\right)} < \frac{k_f\gamma}{\lambda\eta} \qquad [B\text{-}5]$$

$$\frac{i_A}{\sigma} = \frac{\Delta\Phi}{L} = \frac{zF(\rho/M)k_f\gamma}{\sigma\left(\dfrac{2\pi}{\lambda}\right)\lambda\eta}\left[\frac{1}{\sqrt{2}} + \frac{RT\left(\dfrac{2\pi}{\lambda}\right)L}{azF\Delta\Phi}\right] \qquad [B\text{-}6]$$

Multiplying both sides by $[zF(\rho/M)\lambda/\beta]$, the left side reduces to the Peclet number Pe

$$Pe = \frac{\Delta\Phi zF(\rho/M)\lambda}{\beta L} = \frac{\rho}{\eta}\left[\frac{z^2F^2\rho k_f\gamma}{\sigma\left(\dfrac{2\pi}{\lambda}\right)M^2}\right]\left[\frac{1}{\sqrt{2}} + \frac{RT\left(\dfrac{2\pi}{\lambda}\right)L}{azF\Delta\Phi}\right] \qquad [B\text{-}7]$$

On the right side, the first term can be written in terms of Sc given by $(\eta/\rho D)$

$$Pe = \frac{\Delta\Phi zF(\rho/M)\lambda}{\beta L} = \frac{1}{Sc}\left[\frac{z^2F^2\rho k_f\gamma}{\sigma\left(\dfrac{2\pi}{\lambda}\right)M^2D}\right]\left[\frac{1}{\sqrt{2}} - \frac{RT\left(\dfrac{2\pi}{\lambda}\right)L}{azF\Delta\Phi}\right] \qquad [B\text{-}8]$$

Therefore, Pe scales as $1/Sc$, which agrees with the slope shown in Fig. 5.

List of Symbols

a	constant in the anodic potential term
A	perturbation amplitude, m
$\mathcal{B}$	electromigration constant
C	dimensionless concentration
c	molar concentration, mol/m^3
D	diffusivity, m^2/s
F	Faraday's constant, 96480 C/mol
$\mathcal{F}$	free-energy functional
J	flux, mol/m^2 s
K_{ii}	gradient energy penalty coefficient for ternary, J/m
L	separation between electrodes, m
L_{diff}	diffusion lengthscale, m
N	force parameter
Pe	Peclet number
R	gas constant, J/mol K
Sc	Schmidt number
t	time, s
T	temperature, K
$\vec{U}$	velocity vector, m/s
V	volume, m^3

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X mole fraction
z number of electrons transferred per mole, mol e^-/mol

Greek

α phase-field gradient penalty coefficient, J/m
β free-energy coefficient, J/m^3
γ interfacial energy, J/m^2
ε interface thickness, m
ϵ permittivity
η viscosity, Pa s
κ mobility, m^5/J s
λ domain width, perturbation wavelength, m
μ chemical potential, J/m^3
σ ionic conductivity, Ω^{-1} m^{-1}
σ_e electronic conductivity, Ω^{-1} m^{-1}
σ_{eff} effective conductivity, Ω^{-1} m^{-1}
ρ_f charge density, C/m^3
ρ/M molar density, mol/m^3
τ timescale, s
Φ electrical potential, J/C
ϕ dimensionless composition
Ψ homogeneous free energy, J/m^3
ω vorticity, s^{-2}

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